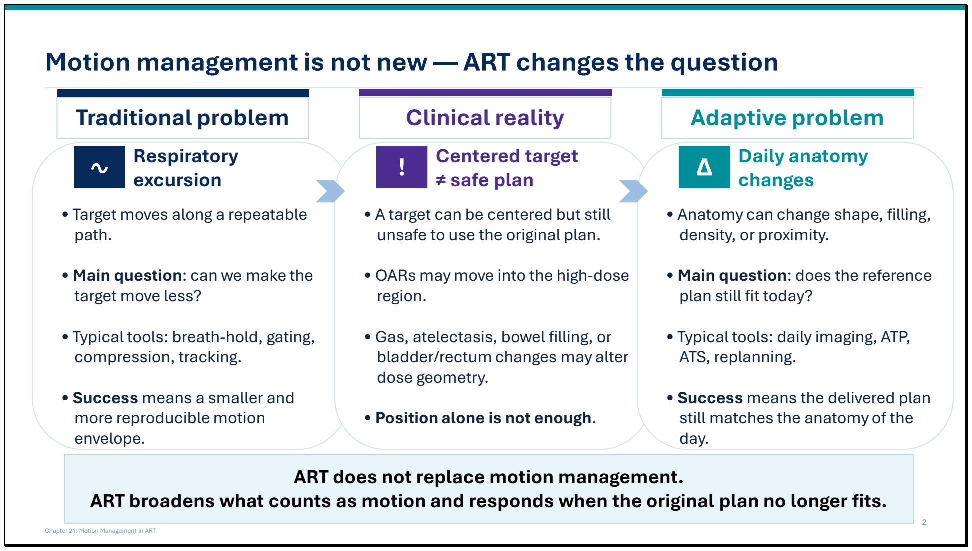
Slide 1

Good morning, everyone. My name is Mihaela Rosu-Bubulac from Virginia Commonwealth University.
Today I will be talking about **motion management in adaptive radiotherapy**, the leap from stabilization to adaptation.
And everything I will discuss today comes back to this idea:
stabilize what moves, recognize what deforms, and adapt when the plan no longer fits.

Slide 2



Motion management is not new.

The **traditional problem** was mainly respiratory excursion, as the target was moving along a repeatable path.
• So the question was: can we make the target move less?
• The typical tools to address this question were breath-hold, gating, compression, and tracking.
• Success in answering the main question meant to have a smaller and more reproducible motion envelope .

But the **clinical reality** is more complicated.
• A target can be centered and still be unsafe to treat with the original plan.
• Organs at risk may move into the high-dose region.
• Gas, atelectasis, bowel filling, or bladder and rectum changes may alter the dose geometry.
• So position alone is not enough.

With all these daily anatomical changes, the question is: does the reference plan still fit today?
New tools now come to play: daily imaging, ATP, ATS, replanning
• Success now takes a new dimension, it means the delivered plan still matches the anatomy of the day.

So ART does not replace motion management.
• It broadens what counts as motion and responds when the original plan no longer fits.

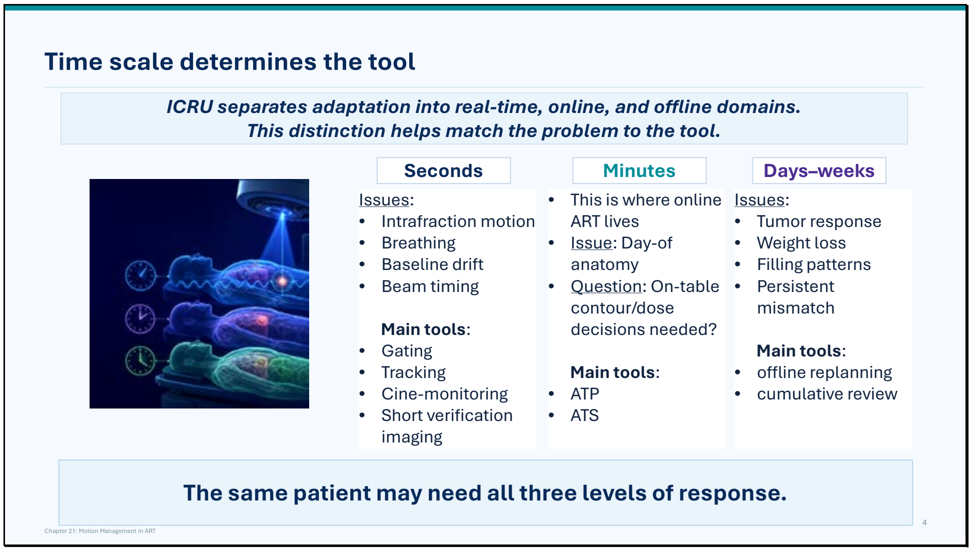
Slide 3

Which time scale is hardest to manage reliably in your clinic?

Raise your hand.

- **A** **Seconds: intrafraction breathing, drift, gating, or tracking**
- **B** **Minutes: online adaptation logistics at the machine**
- **C** **Days to weeks: cumulative anatomy change and replanning**
- **D** **Site-dependent: different answer for lung, liver, pancreas, and prostate**

Slide 4



ICRU separates adaptation into **real-time, online, and offline** domains.
That distinction is useful because it helps us match the problem to the right response.

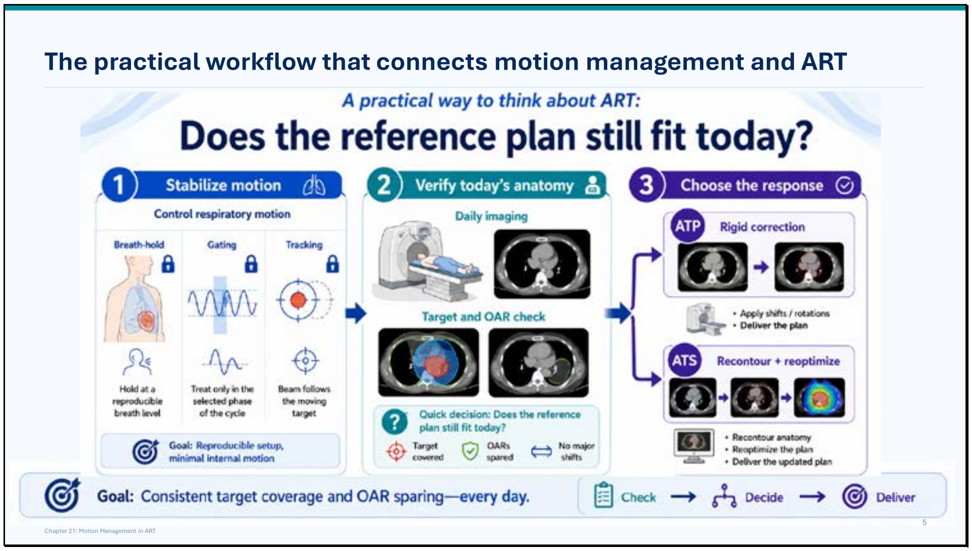
The shortest time scale we are dealing with is **seconds**.
The **issues** are intrafraction motion, breathing, baseline drift, and beam timing.
The **main tools** are gating, tracking, cine monitoring, and short verification imaging.

Then we move to the scale of **minutes**.
This is where online ART lives.
Now the **issue** is the day-of anatomy, and the question is whether we need on-table contour or dose decisions.
The **main tools** here are ATP or ATS at the treatment unit.

Finally, there are changes that occur over **days to weeks**.
This **includes** tumor response, weight loss, changing filling patterns, and persistent mismatch from the original plan.
The main tools here are offline replanning and cumulative review.

So the bottom line is that the same patient may need all three levels of response.

Slide 5



Let's dwell into the practical workflow that connects motion management to adaptive treatment decisions.
The practical way to think about ART is this: **does the reference plan still fit today?**

The first step is to **stabilize motion**.
This can be done with traditional methods:
- breath-hold
- gating
- tracking.

The goal is to have a reproducible setup and minimal internal motion.

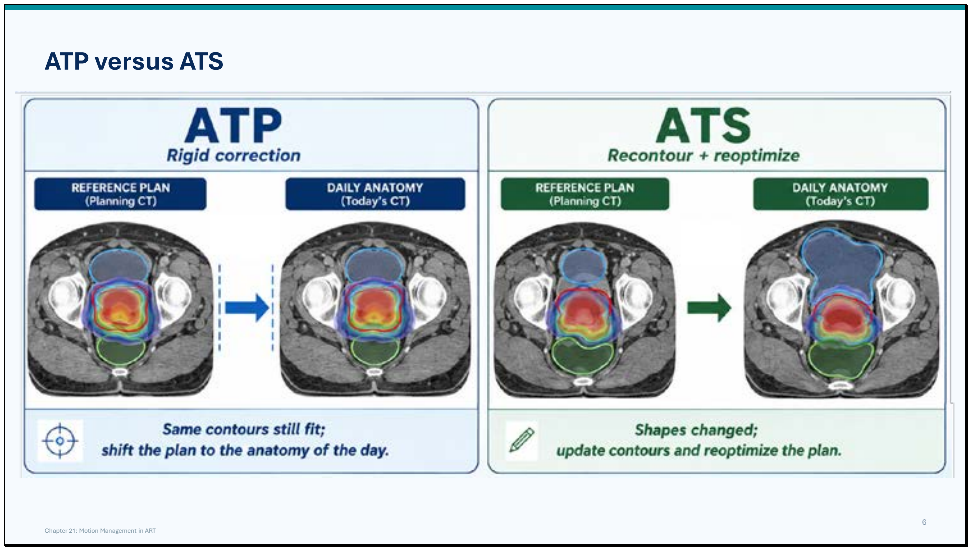
The second step is to **verify today's anatomy**.
Daily imaging is not only for setup but also used to check whether the target is covered, OARs are still spared and whether there are major shifts or geometry changes.

Then the third step is to **choose the response**.
If the anatomy is essentially the same, but shifted, then **ATP** may be enough.
- We apply shifts or rotations and deliver the plan.

But **if the anatomy has changed shape**, **or if the target–OAR relationship has changed**, then **ATS** is needed.
That means recontouring the anatomy, reoptimizing the plan, and delivering an updated plan.

So, to summarize, the **goal** is consistent target coverage and OAR sparing on the anatomy of the day**.**
... and the workflow is: **check, decide, deliver**.

Slide 6

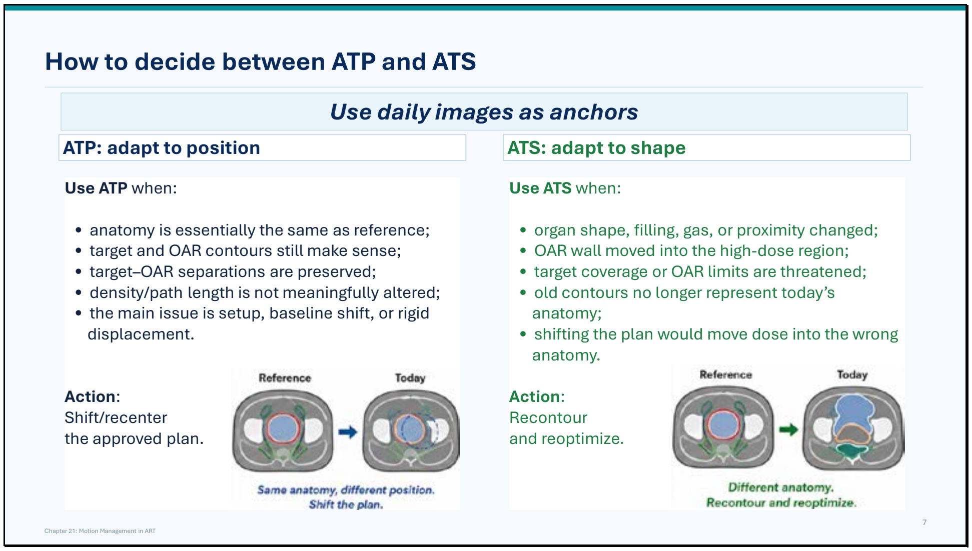


Let's discuss a bit the two adaptive responses: **ATP** and **ATS**.

On the left is **ATP**, or **adapt-to-position**.
ATP is a **rigid correction**.
→ Reference plan and the contours still fit.
Then, we can **shift the plan to the anatomy of the day.**

On the right is **ATS**, or **adapt-to-shape**.
ATS means **recontour and reoptimize**.
Then we **update contours and reoptimize the plan.**

Slide 7



Now that we know what ATS and ATP mean, let's see how we decide which to use.
- We use daily images as our anchors.

We use ATP when:
• anatomy is essentially the same as reference;
• target and OAR contours still make sense;
• target–OAR separations are preserved;
• density/path length is not meaningfully altered;
• the main issue is setup, baseline shift, or rigid displacement.

And the **action** is:
shift/recenter the approved plan.

We use ATS when:
• organ shape, filling, gas, or proximity changed;
• OAR wall moved into the high-dose region;
• target coverage or OAR limits are threatened;
• old contours no longer represent today's anatomy;
• shifting the plan would move dose into the wrong anatomy.
And the **action** is:
recontour and reoptimize.

In short:
- similar anatomy, new position—use ATP.
- new anatomy—use ATS.

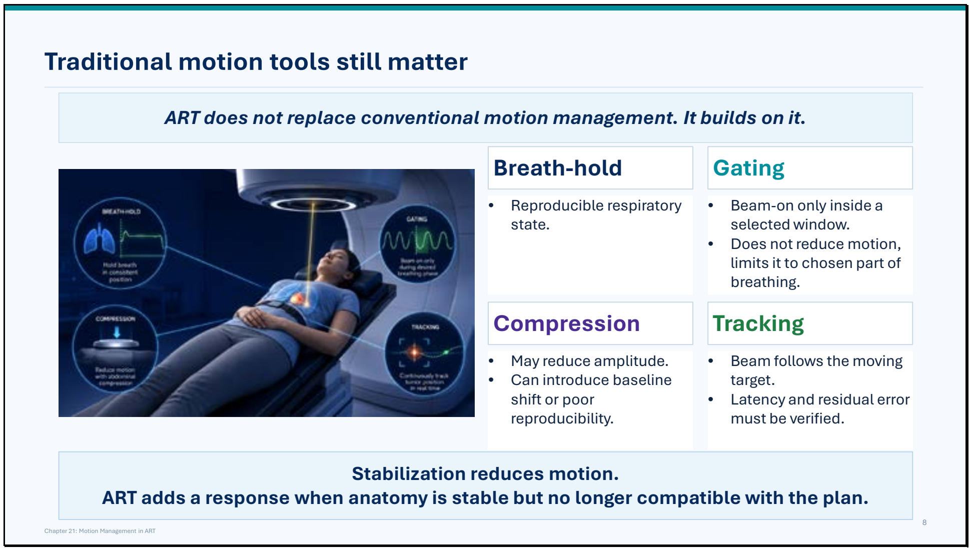
Slide 8

What pushes a case from ATP to ATS in your practice?

Raise your hand.

- **A** **Target coverage concern**
- **B** **OAR proximity or hot spot**
- **C** **Contour uncertainty on the daily image**
- **D** **Time, staffing, QA, or throughput constraints**

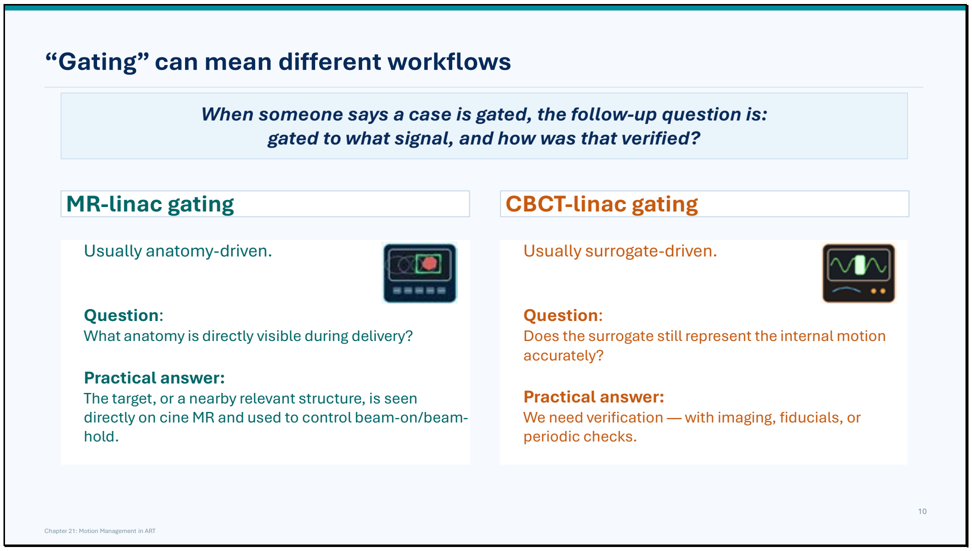
Slide 9



Traditional motion tools still matter.
ART does not replace conventional motion management. It builds on it.
These tools are still the first layer of control, especially when the dominant problem is respiratory motion.
First, **breath-hold**.
Breath-hold gives a reproducible respiratory state.
Second, **gating**.
With gating, the beam is on only inside a selected window.
So gating does not remove motion, but it limits treatment to the part of the breathing cycle we choose.
Third, **compression**.
Compression may reduce motion amplitude.
But it can also introduce baseline shift or poor reproducibility.
Fourth, **tracking**.
With tracking, the beam follows the moving target.
However, latency and residual error must be verified.

So the bottom line is this:
Stabilization reduces motion. ART adds a response when anatomy is stable but no longer compatible with the plan.

Slide 10

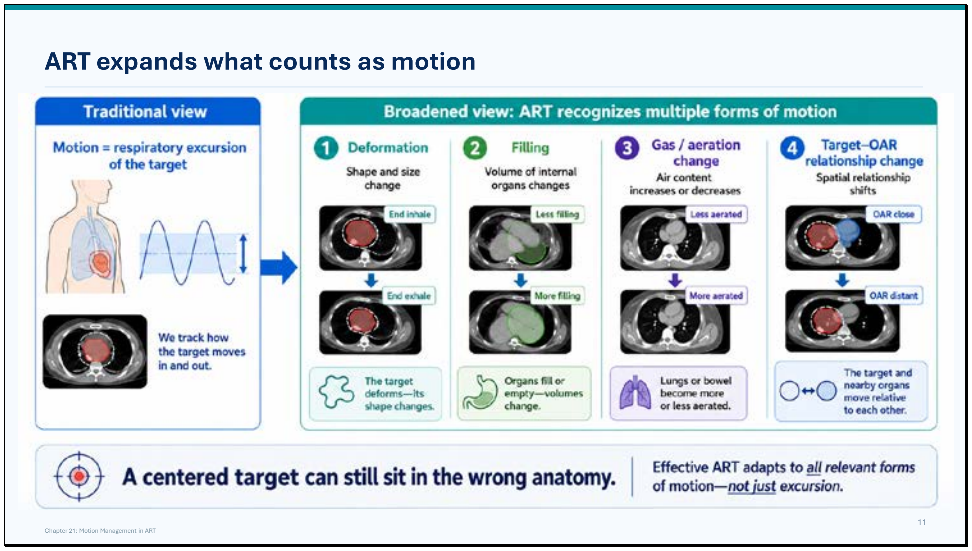


Gating can mean different workflows.
So when someone says a case is gated, the follow-up question should be:
gated to what signal, and how was that verified?

On an **MR-linac**, gating is usually **anatomy-driven**.
The question is:
What anatomy is directly visible during delivery?
The practical answer is that the target, or a nearby relevant structure, is seen directly on cine MR and can be used to control beam-on and beam-hold.
So in this setting, gating is tied closely to the anatomy we are treating.

On a **CBCT-linac**, gating is usually **surrogate-driven**.
The question is:
Does the surrogate still represent the internal motion accurately?
The practical answer is that we need verification — with imaging, fiducials, or periodic checks.

Slide 11



AS mentioned earlier, ART expands what counts as motion.
In the traditional view, motion often meant **respiratory excursion of the target**.
In other words, we tracked how the target moved in and out with breathing.

ART, on the other hand, recognizes multiple forms of motion.
First, there is **deformation**.
The target or nearby anatomy may change shape and size, not just position.
Second, there is **filling**.
Internal organs may fill or empty, so their volumes change.
Third, there is **gas or aeration change**.
The lungs or bowel may become more or less aerated.
Fourth, there is **target–OAR relationship change**.
The target and nearby organs at risk may move relative to each other.

As a consequence, **a centered target can still sit in the wrong anatomy.**
That means effective ART has to adapt to all relevant forms of motion, not just respiratory excursion.

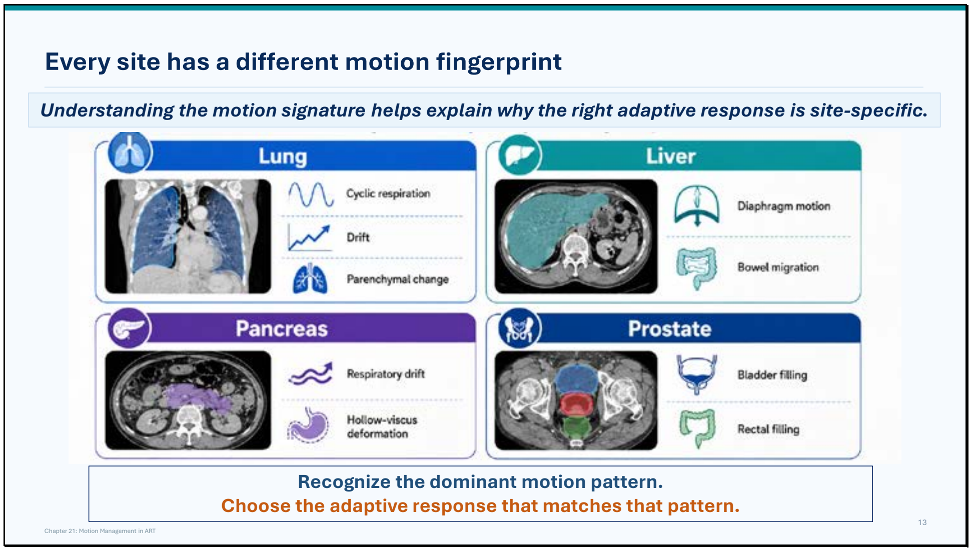
Slide 12

What makes you most comfortable trusting an adapted plan?

Raise your hand.

- **A** **Daily image quality and visibility of relevant anatomy**
- **B** **Contour review process and who signs off**
- **C** **Dose calculation / electron-density workflow**
- **D** **Fast, standardized QA and documentation process**

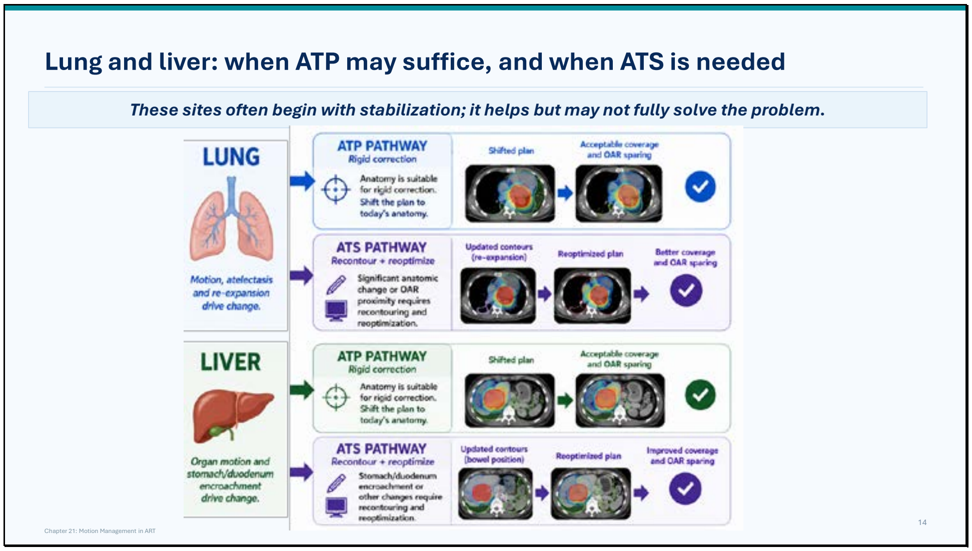
Slide 13



Every site has a different motion fingerprint.
For **lung**, the main issues are cyclic respiration, baseline drift, and parenchymal change.
For **liver**, the motion fingerprint is different.
Respiratory drift is one concern, but so is the deformation of the nearby hollow-viscus,
For **pancreas**, the issue is often not only respiratory drift, but also the nearby bowel migration.
For **prostate**, respiratory motion is not the main concern.
Instead, bladder filling and rectal filling drive the daily anatomy.

So, the bottom line is:
First, recognize the dominant motion pattern.
Next, choose the adaptive response that matches that pattern.

Slide 14



For lung and liver, motion management usually begins with stabilization.
This helps, but may not fully solve the problem.

Starting with **lung**:

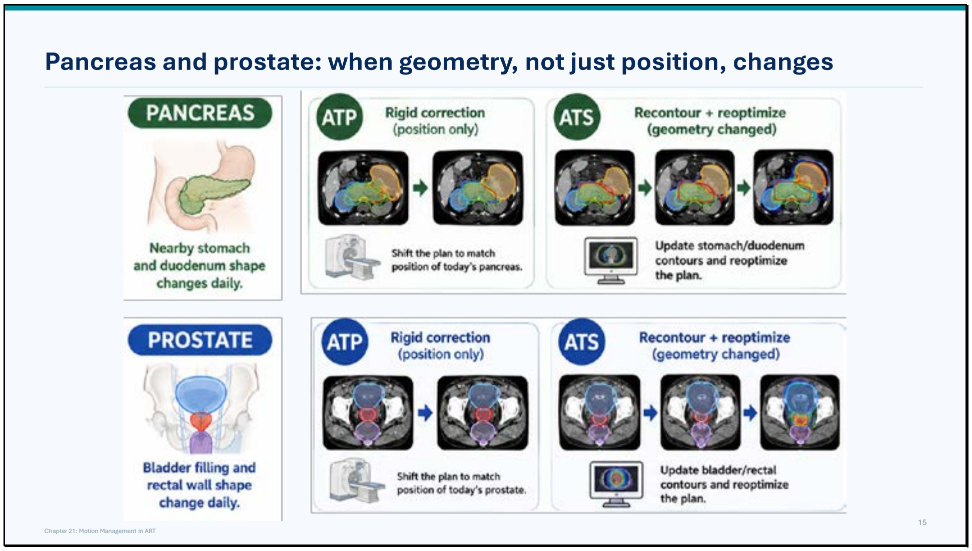
In lung, motion, atelectasis, and re-expansion can drive change.
If the anatomy is still suitable for rigid correction, then the **ATP pathway** may be enough.
In that case, we shift the plan to today's anatomy.
If coverage and OAR sparing remain acceptable, then ATP has done its job.

But if there is significant anatomic change, or if OAR proximity changes, then the **ATS pathway** is needed.
In that case, we update the contours, reoptimize the plan, and aim for better coverage and OAR sparing.

Now for **liver**:
In liver, the issue is organ motion, but also stomach and duodenum position.
If the liver anatomy is still suitable for rigid correction, then **ATP** may be enough.
Again, we shift the plan to today's anatomy.
If coverage and OAR sparing are acceptable, then we can proceed.

But if stomach or duodenum encroachment changes the geometry, then ATP is not enough.
That is when ATS becomes the better pathway.
We update the bowel or stomach and duodenum position, reoptimize the plan, and improve coverage and OAR sparing.

Slide 15

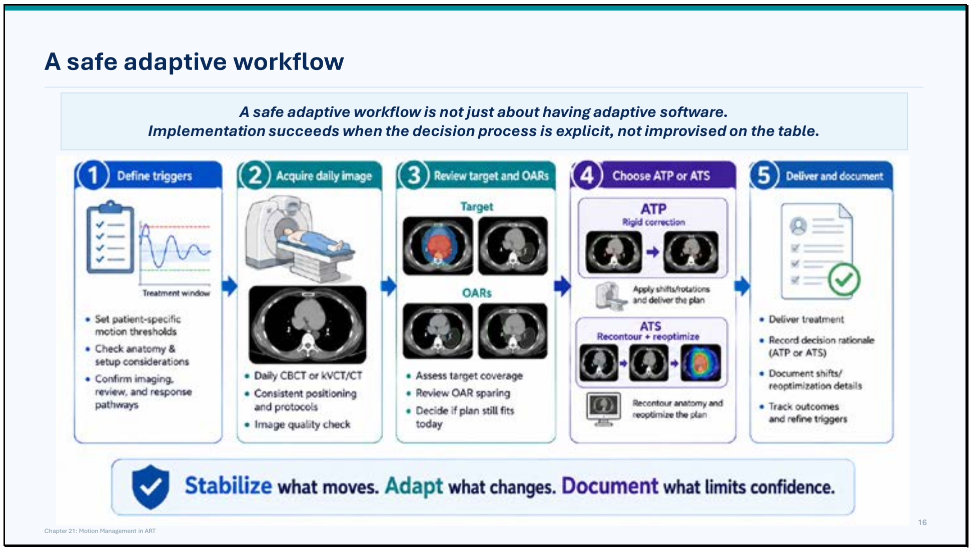


Here, the main message is that **geometry, not just position, changes**.

For **pancreas**, the nearby stomach and duodenum can change shape daily.
- If the change is mainly positional, the **ATP pathway** can be used.
That means rigid correction: shift the plan to match the position of today's pancreas.
- But if the geometry has changed, then **ATS** is needed.
For pancreas, that means updating the stomach and duodenum contours and reoptimizing the plan.

For **prostate**, the same logic applies, but the driving anatomy is different.
- Here, bladder filling and rectal wall shape can change daily.
If the prostate position has changed but the anatomy is otherwise similar, ATP can be used.
- But if bladder or rectal geometry has changed, then ATS is needed.
In that case, we update the bladder and rectal contours and reoptimize the plan.

Slide 16



A safe adaptive workflow is not just about having adaptive software.
The key point is that **implementation succeeds when the decision process is explicit, not improvised on the table.**

The first step is to **define triggers**. That includes patient-specific motion thresholds, anatomy and setup considerations, and the imaging, review, and response pathways.

The second step is to **acquire the daily image**. This may be daily CBCT or kVCT/CT, depending on the platform.
The image has to be acquired with consistent positioning and protocols, and it needs an image quality check before it is used for decisions.

The third step is to **review the target and OARs**.
We ask: is the target still covered?
Are the OARs still spared?
And does the plan still fit today?

The fourth step is to **choose ATP or ATS**.
If the anatomy is suitable for rigid correction, we use ATP: apply shifts or rotations and deliver the plan.
If the anatomy has changed shape or the relationship between target and OARs has changed, we use ATS: recontour the anatomy and reoptimize the plan.

The fifth step is to **deliver and document**.
We deliver the treatment, record the rationale for ATP or ATS, document shifts or reoptimization details, and track outcomes so the triggers can be refined over time.

So the bottom line is:
Stabilize what moves.
Adapt what changes.
Document what limits confidence.

Slide 17

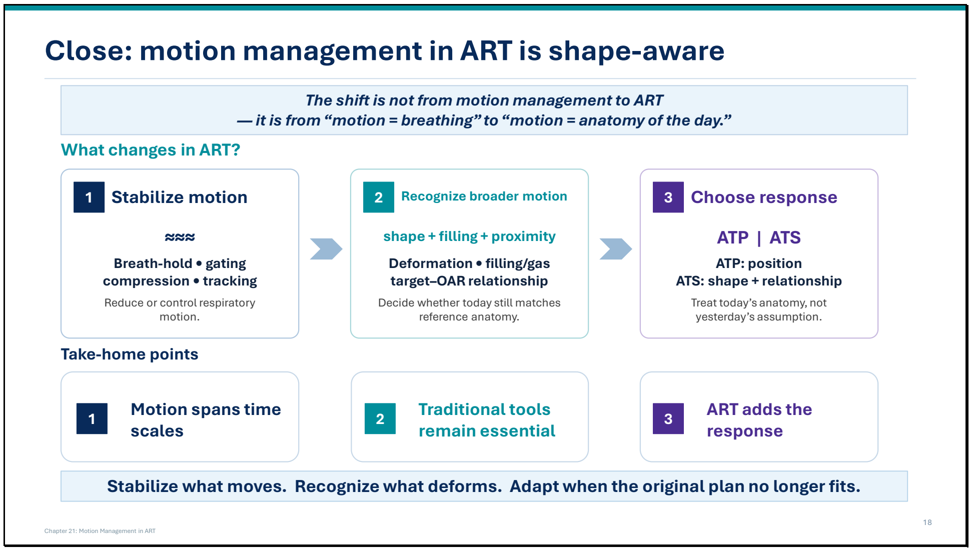
Audience question

If you were expanding ART tomorrow, where would you start?

Raise your hand.

- **A** **Biggest clinical need**
- **B** **Highest patient volume**
- **C** **Best imaging and contour confidence**
- **D** **Best staffing, QA, and implementation readiness**

Slide 18



To close, let me just say this: motion management in ART is **shape-aware**.
The shift is not from motion management to ART.
It is from **motion equals breathing** to **motion equals anatomy of the day**.

So what changes in ART?
- First, we still **stabilize motion**.
Breath-hold, gating, compression, and tracking remain important because they reduce or control respiratory motion.
- Second, we **recognize broader motion**.
In ART, motion also includes shape change, filling, gas, and proximity.
That means we pay attention to deformation, filling and gas, and changes in the target–OAR relationship, to decide whether today still matches the reference anatomy.
Third, we **choose the response**.
If the issue is position, we may use **ATP**.
If the issue is shape or relationship, we may use **ATS**.
The goal is to treat today’s anatomy, not yesterday’s assumption.

The **take-home points** are these:
Motion spans time scales.
Traditional tools remain essential.
And ART adds the response.

So the final message is:
Stabilize what moves.
Recognize what deforms.
Adapt when the original plan no longer fits.